

Heuristic Strategies for Accelerating Recursive Reliability Evaluation of Multistate Networks Using Minimal Path Vectors

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Abstract—Multistate flow networks model real engineering systems in which each link may operate at several capacity levels. The two-terminal reliability of such a network is defined as the probability that the maximum flow from a source to a sink satisfies a required demand level d . A standard exact approach expresses this reliability using the set of minimal path vectors (MPVs), but direct computation via inclusion–exclusion becomes impractical when the number of MPVs is large.

In this paper, we present a recursive reliability evaluation method based on the sum of disjoint products principle and component-wise maximum operations. We also investigate heuristic strategies to improve practical performance by reducing the size of intermediate vector sets through splitting and pruning techniques. Experimental results on randomly generated MPV instances show that suitable splitting heuristics can significantly reduce recursion depth and execution time, enabling efficient exact reliability computation for larger MPV sets.

Index Terms—Multistate network reliability, minimal path vectors, two-terminal reliability, recursive sum of disjoint products, heuristic splitting strategies, K-means clustering.

I. INTRODUCTION

Reliability evaluation is essential for assessing the correct operation of complex engineering systems under uncertainty. Many real-world systems, including communication networks, transportation grids, power systems, IIoT sensor networks, and large-scale computing infrastructures, are subject to partial failures and performance degradation that can affect overall functionality.

Multistate models generalize classical two-state (working/failure) representations by allowing components to operate at multiple discrete performance levels, providing more realistic system descriptions [1], [2]. In this context, the two-terminal reliability problem measures the probability that a system satisfies a required demand level d , such as delivering at least d units of flow between a source and a sink.

Exact reliability evaluation often relies on minimal structural representations. Minimal path vectors (MPVs) and minimal cut vectors (MCVs) capture the smallest configurations that ensure system success or failure, respectively. These representations are widely used to describe the system success set [3], [4], and various algorithms have been proposed for their enumeration and for reliability computation based on them [5]–[7].

Despite extensive research on combinatorial and structural methods, reliability computation remains challenging for large systems due to the exponential growth in the number of MPVs and MCVs. The inclusion–exclusion principle becomes computationally infeasible in such cases. Recursive decomposition approaches, such as the Recursive Sum of Disjoint Products (RSDP) algorithm introduced by Zuo *et al.* [9], address this issue by reducing redundancy through systematic recursion.

In this paper, we investigate whether further computational improvements can be achieved by introducing an intelligent partitioning strategy for minimal path vectors prior to recursive reliability evaluation. Our hypothesis is that grouping structurally similar MPVs increases the likelihood of dominated vectors during intersection operations, thereby enhancing pruning efficiency, reducing the number of recursive calls, and improving overall performance.

The novelty of this work lies in extending the classical recursive Sum of Disjoint Products (SDP) reliability evaluation framework with an additional heuristic decision layer for MPV partitioning. In addition to evaluating distance-based and clustering-based partition strategies, we propose a classifier-guided selection mechanism that determines the most suitable partitioning strategy for a given MPV set before recursive decomposition begins. This adaptive mechanism leads to different recursive execution paths, influences dominated-vector

pruning behavior, and results in measurable variations in computational efficiency. Thus, the contribution of this paper is not only an empirical comparison of existing heuristics, but also a new algorithmic framework for adaptive control of recursive reliability evaluation.

II. RELATED WORK

The study of reliability in complex systems spans several decades and encompasses analytical models, combinatorial enumeration, and algorithmic decomposition techniques. Early foundational work on coherent multistate system reliability established structural representations for component and system states [1], [2]. In flow network reliability, minimal path and minimal cut representations have become standard modeling tools, as they provide compact and effective characterizations of system success and failure events [3], [4].

The enumeration of minimal path vector (MPV) and minimal cut vector (MCV) sets has been the subject of several algorithms tailored to multi-state systems with capacity constraints [5]–[7]. Once these sets are obtained, exact reliability can be computed using either the inclusion-exclusion principle or the Sum of Disjoint Products (SDP) method [8]. However, the inclusion-exclusion approach suffers from exponential growth in the number of terms as the MPV set increases, making it impractical for large instances. This limitation has motivated the development of recursive SDP-based methods that exploit structural properties to reduce redundant computations. Among these, the Recursive Sum of Disjoint Products (RSDP) algorithm proposed by Zuo *et al.* [9] is one of the most effective, providing systematic decomposition and vector simplification for large MPV sets.

Beyond purely combinatorial enumeration, recent studies have proposed enhanced strategies for improving reliability evaluation in complex systems. Abdullah *et al.* [10] introduce an analytical approach for computing minimal path and cut sets that emphasizes efficient connectivity analysis and simplified procedures. In sensor-based IIoT networks [11], minimal cut set-based reliability estimation (MCRel) techniques have also been developed, demonstrating improved performance over alternative methods, particularly under high network load conditions.

In a different domain, Kokolis *et al.* [12] investigate infrastructure reliability in large-scale machine learning clusters, focusing on failure taxonomies, job failure dynamics, and operational reliability metrics across extensive GPU workloads. Their findings highlight important reliability challenges in modern AI training environments, where system reliability directly affects both performance and resource utilization.

Despite these advances, exact evaluation of reliability remains computationally challenging for large minimal path vector sets. Consequently, methods that reduce intermediate computation, such as dominated vector pruning and heuristic splitting strategies, continue to be active areas of research. Motivated by these considerations, this research investigates heuristic strategies for partitioning MPV sets prior to recur-

sive SDP computation, with the goal of maximizing pruning efficiency and minimizing runtime overhead.

III. PRELIMINARIES

Consider a multistate system consisting of m independent components. Component i can operate in $M_i + 1$ discrete and mutually exclusive states:

$$0, 1, \dots, M_i,$$

where 0 denotes failure and larger values indicate higher capacity. The system state is represented by the vector

$$\mathbf{x} = (x_1, x_2, \dots, x_m).$$

A. State Function and Reliability

Let

$$\phi : \{0, \dots, M_1\} \times \dots \times \{0, \dots, M_m\} \rightarrow \{0, \dots, M\}$$

denote the system performance function. For a required demand level d , the reliability at level d is defined as

$$R_d = \Pr(\phi(\mathbf{X}) \geq d). \quad (1)$$

For two vectors $\mathbf{x}$ and $\mathbf{y}$ of equal length, we write $\mathbf{x} \geq \mathbf{y}$ if $x_i \geq y_i$ for all components i . Similarly, $\mathbf{x} > \mathbf{y}$ denotes $\mathbf{x} \geq \mathbf{y}$ and $\mathbf{x} \neq \mathbf{y}$.

B. Minimal Path Vectors

A vector $\mathbf{x}$ is called a *path vector* to level d if $\phi(\mathbf{x}) \geq d$, and a *cut vector* to level d if $\phi(\mathbf{x}) < d$.

Definition 1. A vector $\mathbf{x}$ is called a *minimal path vector* (MPV) to level d if:

- $\phi(\mathbf{x}) \geq d$, and
- for every $\mathbf{y} < \mathbf{x}$, we have $\phi(\mathbf{y}) < d$.

Let the complete set of minimal path vectors be denoted by

$$\mathcal{P} = \{\mathbf{p}^{(1)}, \mathbf{p}^{(2)}, \dots, \mathbf{p}^{(L)}\}.$$

Then the event $\{\phi(\mathbf{X}) \geq d\}$ can be expressed as the union

$$\{\phi(\mathbf{X}) \geq d\} = \bigcup_{i=1}^L \{\mathbf{X} \geq \mathbf{p}^{(i)}\}. \quad (2)$$

C. Component Probabilities

Let $p_{i,j} = \Pr(X_i = j)$ denote the probability that component i is in state j . We define the tail probability as

$$P_{i,k} = \Pr(X_i \geq k) = \sum_{j=k}^{M_i} p_{i,j}. \quad (3)$$

Assuming independence of component states, for any MPV $\mathbf{p}$ we have

$$\Pr(\mathbf{X} \geq \mathbf{p}) = \prod_{i=1}^m P_{i,p_i}. \quad (4)$$

IV. RECURSIVE SDP RELIABILITY EVALUATION

In this section, we present the recursive algorithm used in our study. The method is based on the SDP principle and assumes that the complete set of minimal path vectors (MPVs) is available.

A. Union Decomposition

Let $\mathcal{P}$ denote the set of MPVs. The reliability can be expressed as

$$R_d = \Pr \left(\bigcup_{\mathbf{p} \in \mathcal{P}} \{\mathbf{X} \geq \mathbf{p}\} \right).$$

To avoid the exponential expansion associated with the inclusion–exclusion principle, we recursively split $\mathcal{P}$ into two disjoint subsets A and B :

$$\mathcal{P} = A \cup B, \quad A \cap B = \emptyset.$$

Using the standard union formula, we then have

$$\Pr(A \cup B) = \Pr(A) + \Pr(B) - \Pr(A \cap B), \quad (5)$$

where $\Pr(A)$ denotes the probability of the union of MPV events indexed by A .

B. Intersection via Componentwise Maximum

Let $\mathbf{a}$ and $\mathbf{b}$ be two MPVs. The intersection of their corresponding events can be expressed as

$$\{\mathbf{X} \geq \mathbf{a}\} \cap \{\mathbf{X} \geq \mathbf{b}\} = \{\mathbf{X} \geq \max(\mathbf{a}, \mathbf{b})\},$$

where the maximum is computed componentwise:

$$(\max(\mathbf{a}, \mathbf{b}))_i = \max(a_i, b_i), \quad i = 1, \dots, m.$$

Extending this to two MPV subsets A and B , the intersection set is generated as

$$A \cap B = \{\max(\mathbf{a}, \mathbf{b}) : \mathbf{a} \in A, \mathbf{b} \in B\}. \quad (6)$$

C. Pruning Dominated Vectors

The intersection in (6) may produce many redundant vectors. A vector $\mathbf{u}$ is said to be *dominated* by $\mathbf{v}$ if

$$u_i \geq v_i \quad \forall i \quad \text{and} \quad \mathbf{u} \neq \mathbf{v}.$$

If $\mathbf{u}$ is dominated by $\mathbf{v}$, then

$$\{\mathbf{X} \geq \mathbf{u}\} \subseteq \{\mathbf{X} \geq \mathbf{v}\},$$

implying that $\mathbf{u}$ does not contribute additional probability mass and can be safely removed.

After each intersection step, we prune dominated vectors to retain only minimal vectors. This step is crucial for practical performance and significantly reduces the size of intermediate sets.

D. Recursive Procedure

The recursive reliability computation can be summarized as follows:

- 1) If the MPV set contains only one vector $\mathbf{p}$, return $\Pr(\mathbf{X} \geq \mathbf{p})$ using (4).
- 2) Split the set into two disjoint subsets A and B .
- 3) Recursively compute $R(A)$ and $R(B)$.
- 4) Construct $A \cap B$ using componentwise maxima.
- 5) Prune dominated vectors from $A \cap B$.
- 6) Recursively compute $R(A \cap B)$.

7) Return $R(A) + R(B) - R(A \cap B)$.

This procedure is exact and terminates after a finite recursion depth. However, its runtime strongly depends on the growth of intermediate sets, which motivates the heuristic splitting strategies described in the next section.

V. HEURISTIC SPLITTING STRATEGIES

The performance of the recursive algorithm strongly depends on how the MPV set is partitioned at each recursion step. Poor splitting may lead to large intersection sets and weak pruning, resulting in excessive recursion. To address this issue, we investigate heuristics for dividing the MPV set into two subsets in a way that increases similarity within each subset and encourages stronger pruning during intersections.

A. Distance-Based Splitting

Given a set of vectors $\mathcal{P}$, we first compute the pairwise distance matrix using the Minkowski metric:

$$d_p(\mathbf{x}, \mathbf{y}) = \left(\sum_{i=1}^m |x_i - y_i|^p \right)^{1/p}, \quad p \geq 1, \quad (7)$$

and for $p = \infty$, the Chebyshev distance is used.

The two vectors with the largest pairwise distance are selected as representatives of two clusters. Each remaining vector is then assigned to the subset corresponding to the closer representative, yielding two subsets A and B .

B. Subset-Minimum Distance Assignment

Instead of assigning each vector based solely on its distance to a single representative, we may assign a vector to the subset for which its minimum distance to any existing element is smaller. This approach tends to produce more compact clusters, potentially increasing pruning efficiency, but results in additional computational cost.

C. K-means Clustering

Alternatively, we can split MPVs using the K-means clustering algorithm with $k = 2$. The resulting clusters define the subsets A and B . Various initialization strategies can be employed, including random initialization or selection of the two most distant vectors as initial centroids.

VI. EXPERIMENTAL EVALUATION

A. Experimental Setup

To evaluate the practical performance of the proposed recursive reliability algorithm, we conducted an experimental comparison between the classical iterative approach based on the Inclusion-Exclusion (IE) principle and several recursive strategies enhanced with pruning of dominated vectors.

The experiments were performed on randomly generated instances of minimal path vectors (MPVs), with input sizes ranging from 10 up to 300 vectors. For each test instance, component state probabilities were generated and the reliability value was computed using both methods whenever possible.

In the experiments presented in the following figures, both the MPV dimension and the number of vectors were varied.

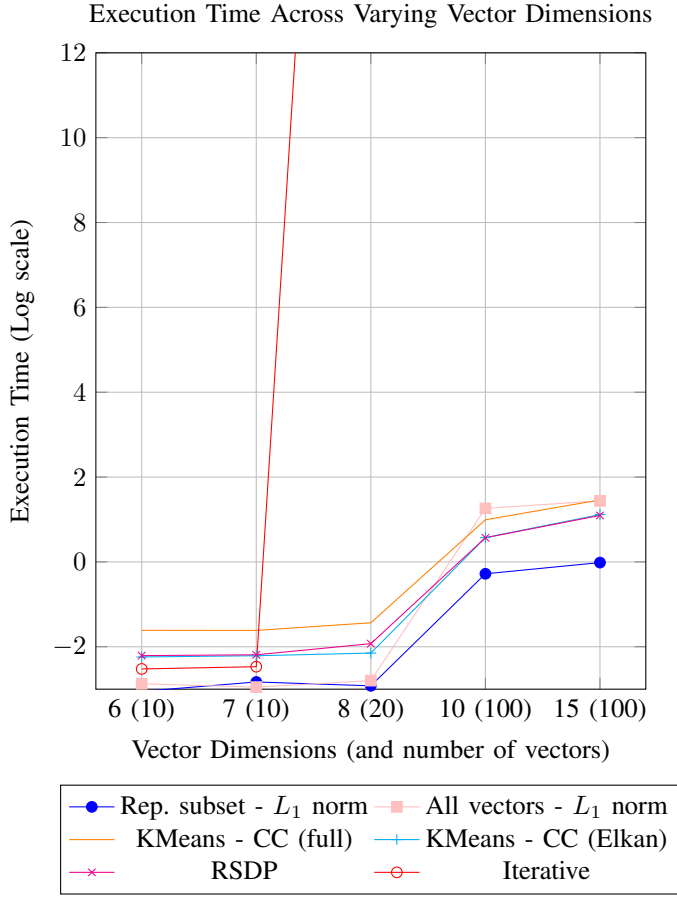


Fig. 1. Execution Time Comparison of Algorithms Across Varying Vector Dimensions

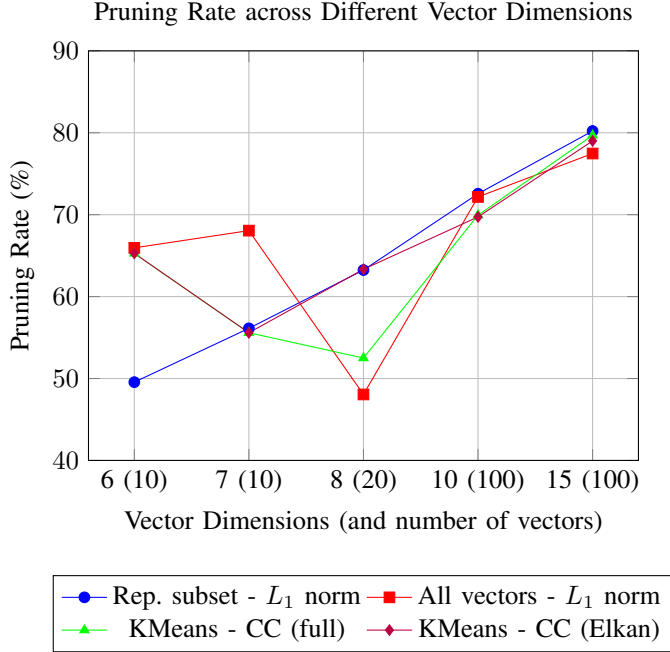


Fig. 2. Pruning Rate across Different Vector Dimensions

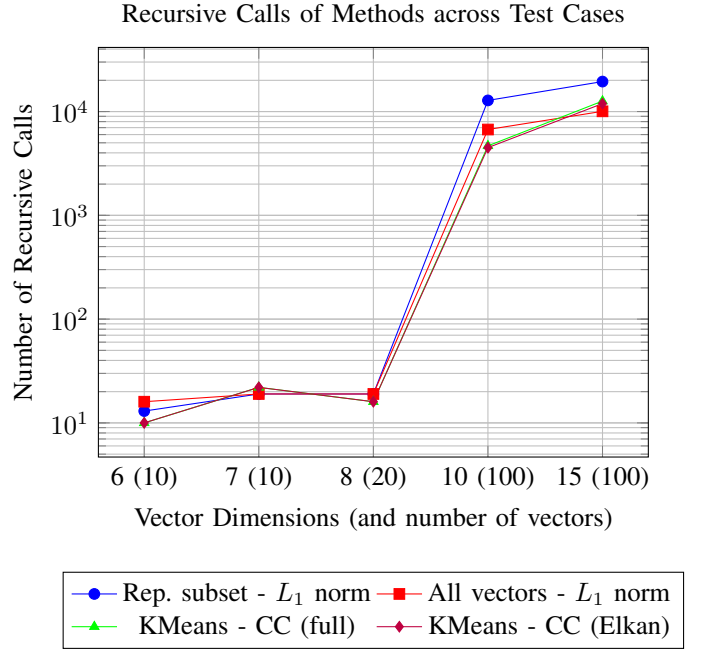


Fig. 3. Number of Recursive Calls for Different Methods and Test Cases

Specifically, dimensions $m \in \{6, 7, 8, 10, 15\}$ were considered, with the corresponding number of MPVs indicated in parentheses in the plots.

In the recursive method, different strategies for splitting the MPV set into two subsets were tested. In particular, we examined splitting based on distance measures between vectors using different Minkowski metrics (Manhattan, Euclidean, and Chebyshev), as well as clustering-based splitting using the K-means algorithm.

The following performance metrics were recorded: execution time, number of recursive calls, and pruning rate (percentage of dominated vectors removed).

B. Evaluation Metrics

The main objective of the recursive method is to reduce the number of intermediate computations by eliminating dominated vectors generated during recursive decomposition. Therefore, in addition to runtime, the pruning rate was considered an important indicator of efficiency. A higher pruning rate typically leads to fewer recursive calls and smaller intermediate sets, which reduces the overall computational effort.

All plots in this section present *mean values* over multiple randomly generated instances for each tested vector dimension.

C. Results and Discussion

The experimental results confirm that the iterative Inclusion-Exclusion method becomes computationally infeasible even for moderate MPV set sizes. In particular, for instances with more than approximately 20 MPVs, the iterative approach required extremely large execution times and was practically unusable.

On the other hand, the recursive method demonstrated significantly improved scalability. Although for small instances the iterative approach can be faster due to lower overhead, the recursive algorithm becomes substantially superior as the number of MPVs increases. This behavior is mainly explained by the pruning of dominated vectors, which prevents exponential blow-up in many practical cases.

Furthermore, the experiments show that the splitting strategy has a strong influence on performance. In most tested cases, K-means clustering (initialized using the farthest pair of vectors) produced the best results in terms of runtime and number of recursive calls. Distance-based splitting using Manhattan (L_1) and Euclidean (L_2) metrics also produced competitive performance, while Chebyshev distance (L_∞) generally performed worse.

Moreover, as shown in Fig. 1, several of the proposed splitting heuristics outperform the classical RSDP method [9] in terms of execution time, particularly for larger MPV sets, demonstrating the practical advantage of heuristic partitioning prior to recursive evaluation.

Interestingly, the pruning rate does not show a clear monotonic dependence on the number of MPVs. Instead, it fluctuates across different instance sizes and strategies. This indicates that pruning effectiveness depends more strongly on the structural properties of the generated MPV sets (i.e., their distribution and dominance relationships) than on the MPV set size alone. In other words, even large MPV families may generate comparable pruning rates if their vectors exhibit similar dominance patterns.

Overall, the results indicate that the recursive approach combined with an appropriate splitting heuristic can make exact reliability computation feasible even for instances containing several hundred minimal path vectors.

D. Summary of Findings

The experiments lead to the following conclusions:

- The iterative Inclusion-Exclusion approach is infeasible for MPV sets larger than approximately 20 vectors.
- The recursive method remains practical for much larger MPV sets due to systematic pruning.
- K-means splitting (initialized by the farthest vector pair) provides the best overall performance.
- Manhattan and Euclidean distance metrics are acceptable alternatives, while Chebyshev distance tends to perform poorly.
- The pruning rate appears to depend primarily on the structure of MPV sets rather than their size, and thus exhibits non-monotonic behavior.

VII. CONCLUSION

In this paper, we addressed the exact reliability evaluation of multistate two-terminal networks given the complete set of minimal path vectors. We implemented a recursive SDP-based algorithm and explored heuristic strategies for partitioning the MPV set to enhance pruning efficiency and reduce computational overhead. Our experimental results demonstrate that

suitable splitting heuristics, particularly those based on distance metrics and clustering, can significantly reduce recursion depth and runtime for large MPV sets.

For future work, we plan to extend this approach to real-world network topologies, evaluate alternative clustering and partitioning techniques, and explore machine learning-based methods for adaptive heuristic selection to further improve computational efficiency in complex network scenarios.

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